

OBS GUI and Config Setting Reference

Generated on 2026-05-16 from x64\DML\config.ini.

Searchable Directory

Use PDF search for any GUI label, config key, tab, section, or source file. The companion CSV at docs/settings-reference/settings-index.csv contains the same searchable directory in row form.

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Capture Tab

Sliders

Section	Slider	Config key	Range	Current	Notes
General Capture	Capture FPS	capture_fps	0-240	70	Limits capture loop FPS. 0 disables the cap path and the GUI warns at 0 or 61+.
Capture Preview	Debug scale	preview_debug_scale	0.1x-2.0x	n/a	Preview-only scale slider. It is not saved to config.ini.
Virtual Camera	Virtual camera width	virtual_camera_width	128-3840	1920	Applies when capture_method is virtual_camera.
Virtual Camera	Virtual camera heigth	virtual_camera_heigth	128-2160	1080	Spelling follows the existing config key. Applies when capture_method is virtual_camera.

Activate/Deactivate

Section	Control	Config key	Values	Current	Notes
General Capture	Circle mask	circle_mask	true/false	true	Applies the circular capture mask.
General Capture	Use CUDA Direct Capture	capture_use_cuda	true/false	n/a	CUDA/TensorRT builds only. Available with duplication_api capture.
Capture Preview	Show Preview Window	show_window	true/false	true	Shows the capture preview/debug frame inside the Capture tab.
WinRT	Capture Borders	capture_borders	true/false	true	WinRT capture only. Disabled automatically on unsupported Windows builds.
WinRT	Capture Cursor	capture_cursor	true/false	true	WinRT capture only. Includes the cursor in captured frames.

Other GUI-Exposed Config Controls

Section	Control	Config key	Type/options	Current	Notes
General Capture	Detection Resolution	detection_resolution	160, 320, 640	640	Changes detector input resolution and restarts/reloads dependent paths.
General Capture	Capture method	capture_method	duplication_api, winrt, virtual_camera, udp_capture	duplication_api	Selects frame source.
WinRT	Capture target (WinRT)	capture_target	monitor, window	monitor	WinRT only.
WinRT	Window title contains	capture_window_title	text		Used when WinRT target is window.
Monitor Capture	Capture monitor	monitor_idx	monitor index	0	Monitor list is built from active displays.
Virtual Camera	Virtual camera	virtual_camera_name	available cameras	None	Filtered list of DirectShow video input devices.
UDP Capture	UDP IP	udp_ip	IPv4/string	0.0.0.0	Applied with the Apply UDP Settings button.
UDP Capture	UDP Port	udp_port	1-65535	1234	Applied with the Apply UDP Settings button.

Target Tab

Sliders

Section	Slider	Config key	Range	Current	Notes
Offsets	Approximate Body Y Offset	body_y_offset	0.00-1.00	0.06	Vertical aim point used for body-class detections.
Offsets	Approximate Head Y Offset	head_y_offset	0.00-1.00	0.38	Vertical aim point used for head-class detections.

Activate/Deactivate

Section	Control	Config key	Values	Current	Notes
Targeting	Disable Headshot	disable_headshot	true/false	false	Prevents the head class from being selected as the aim target.
Targeting	Auto Aim	auto_aim	true/false	false	Allows target selection/movement logic to act automatically while held conditions are met.

Mouse Tab

Sliders

Section	Slider	Config key	Range	Current	Notes
FOV	FOV X	fovX	10-120	70	Horizontal field of view for the mouse controller.
FOV	FOV Y	fovY	10-120	70	Vertical field of view for the mouse controller.
State Estimator	Acquisition frames	kalman_acquisition_frames	3-5	n/a	Frames used to ramp prediction weight after target acquisition.
State Estimator	Process noise position	kalman_process_noise_position	0.0001-5000	n/a	Position process noise. Higher values adapt faster to position changes.
State Estimator	Process noise velocity	kalman_process_noise_velocity	0.0001-50000	n/a	Velocity process noise. Higher values adapt faster to speed changes.
State Estimator	Measurement noise	kalman_measurement_noise	0.0001-5000	n/a	Measurement noise. Higher values trust detections less.
State Estimator	Velocity damping	kalman_velocity_damping	0.0-3.0	n/a	Damps velocity estimates to reduce runaway prediction.
State Estimator	Max velocity (px/s)	kalman_max_velocity	100-60000	n/a	Maximum allowed tracked velocity.
State Estimator	Warmup frames	kalman_warmup_frames	0-20	n/a	Legacy warmup frames when latency sweep is off.
State Estimator	Additional prediction (ms)	kalman_additional_prediction_ms	-80.0-120.0	n/a	Manual extra prediction offset in milliseconds.
State Estimator	Reset timeout (s)	kalman_reset_timeout_sec	0.05-3.0	n/a	Time without stable observations before estimator state resets.
Pure PID Movement	Actuator Hz	pid_actuator_hz	30-2000	2000	PID update rate used by the mouse actuator.
Pure PID Movement	Kp	pid_kp	0.0000-1.5000	0.0115	Proportional PID gain.
Pure PID Movement	Ki	pid_ki	0.0000-0.5000	0.0003	Integral PID gain.
Pure PID Movement	Kd	pid_kd	0.0000-0.2500	0.0001	Derivative PID gain.
Pure PID Movement	Deadzone (px)	pid_deadzone_px	0.000-10.000	1.500	Minimum error radius before movement is applied.
Pure PID Movement	Max step (px/tick)	pid_max_pixel_step	0.010-20.000	20.000	Maximum movement step per PID tick from the GUI. Runtime clamp allows up to 80.
Pure PID Movement	Output scale	pid_output_scale	0.010-3.000	0.209	Base multiplier applied to PID output.
Pure PID Movement	Min output scale	pid_min_output_scale	0.000-3.000	0.100	Lower bound for adaptive output scaling.
Pure PID Movement	Max output scale	pid_max_output_scale	0.010-3.000	0.509	Upper bound for adaptive output scaling.
Pure PID Movement	Size reference (px)	pid_size_reference_px	1.0-240.0	48.000	Target size used as the neutral reference for size scaling.
Pure PID Movement	Small target scale	pid_size_min_scale	0.010-1.000	0.200	Minimum scale when targets are smaller than the reference.
Pure PID Movement	Large target scale	pid_size_max_scale	0.050-2.000	1.000	Maximum scale when targets are larger than the reference.
Pure PID Movement	Precision radius / size	pid_precision_radius_scale	0.0000-0.1000	0.020	Target-size-relative radius where movement is considered precise enough.
Pure PID Movement	Slowdown radius / size	pid_slowdown_radius_scale	0.010-1.000	0.300	Target-size-relative radius where scaling slows approach.
Pure PID Movement	Overshoot brake	pid_overshoot_brake	0.010-1.000	0.350	Reduces movement after crossing past the target.
Pure PID Movement	Divergence boost	pid_divergence_boost	0.000-2.000	0.350	Boosts output when the target error is growing.
Pure PID Movement	Scale response	pid_scale_response	0.1-40.0	8.000	Responsiveness of adaptive output scale changes.
Pure PID Movement	Max integral	pid_max_integral	0.0-10000.0	120.000	Caps integral accumulation.
Pure PID Movement	Max derivative term	pid_max_derivative_term	0.000-5.000	0.020	Caps derivative contribution from a single axis.
Pure PID Movement	Derivative filter (ms)	pid_derivative_filter_tau_ms	0.0-250.0	18.000	Smoothing time constant for derivative filtering.
Pure PID Movement	Target timeout (ms)	pid_target_loss_timeout_ms	10.0-1000.0	90.000	Time without target observation before PID state resets.
Pure PID Movement	Feed-forward gain	pid_feed_forward_gain	0.000-4.000	0.201	Strength of feed-forward movement.
Pure PID Movement	Feed-forward lookahead (ms)	pid_feed_forward_lookahead_ms	0.0-120.0	16.800	Prediction lookahead horizon.
Pure PID Movement	Feed-forward frame lookahead	pid_feed_forward_frame_lookahead	0-2	n/a	Additional frame-based velocity lead used when latency sweep is enabled.
Pure PID Movement	Feed-forward max step (px/tick)	pid_feed_forward_max_step	0.000-5.000	5.000	Caps feed-forward contribution per tick.
Pure PID Movement	Feed-forward min speed (px/s)	pid_feed_forward_min_speed	0.0-3000.0	3000.000	Minimum target speed required before feed-forward contributes.
Pure PID Movement	Feed-forward confidence floor	pid_feed_forward_confidence_floor	0.000-1.000	0.527	Minimum observation confidence for feed-forward contribution.
Pure PID Movement	Integration error limit (px)	pid_conditional_integration_error_px	0.0-240.0	n/a	Maximum per-axis error allowed for conditional integral accumulation.
Pure PID Movement	Adaptive error scale (px)	pid_adaptive_output_error_scale	1.0-640.0	n/a	Error distance used as the adaptive output scaling reference.

Section	Slider	Config key	Range	Current	Notes
Pure PID Movement	Derivative smoothing multiplier	pid_derivative_smoothing_multiplier	1.00-6.00	n/a	Multiplies derivative filter tau when latency sweep is enabled.
Pure PID Movement	Governor blend	pid_governor_blend	0.00-1.00	0.200	Blend strength for governor-generated PID scales.
Pure PID Movement	Governor max speed multiple	pid_governor_max_speed_multiple	1.00-5.00	1.000	Maximum speed multiplier allowed when governor output is active.
Game Profile	Sensitivity	Games.<profile>.sens	0.0010-10.0000	profile/local	Editable for custom profiles. UNIFIED is shown read-only.
Game Profile	Yaw	Games.<profile>.yaw	0.0010-0.1000	profile/local	Horizontal degree-to-count conversion for a custom profile.
Game Profile	Pitch	Games.<profile>.pitch	0.0010-0.1000	profile/local	Vertical degree-to-count conversion for a custom profile.
Game Profile	Base FOV	Games.<profile>.baseFOV	10.0-180.0	profile/local	Shown only when FOV Scaled is enabled on a custom profile.
Auto Shoot	bScope Multiplier	bScope_multiplier	0.5-2.0	1.0	Multiplier used by auto-shoot scope timing/behavior. Disabled in UI until Auto Shoot is on.

Activate/Deactivate

Section	Control	Config key	Values	Current	Notes
State Estimator	Runtime latency sweep	runtime_latency_sweep_enabled	true/false	n/a	Enables side-by-side latency sweep behavior for estimator and PID experiments.
State Estimator	Enable Kalman estimator	kalman_enabled	true/false	n/a	Enables Kalman filtering for target motion estimation.
State Estimator	Seed velocity on acquire	kalman_velocity_seed_enabled	true/false	n/a	Seeds velocity from early measurement deltas during acquisition.
State Estimator	Compensate detection delay	kalman_compensate_detection_delay	true/false	n/a	Accounts for detector latency in prediction.
Pure PID Movement	Feed-forward prediction	pid_feed_forward_enabled	true/false	true	Adds velocity-based prediction ahead of pure PID output.
Pure PID Movement	Conditional integration	pid_conditional_integration_enabled	true/false	n/a	Prevents integral windup when latency sweep is enabled and error/output conditions are unsafe.
Pure PID Movement	Adaptive output scaling	pid_adaptive_output_scaling_enabled	true/false	n/a	Enables error-magnitude output scaling when latency sweep is enabled.
Pure PID Movement	Enable PID governor	pid_governor_enabled	true/false	true	Enables the ONNX PID governor when a compatible model can be loaded.
Game Profile	FOV Scaled	Games.<profile>.fovScaled	true/false	profile/local	When enabled, the profile also uses a Base FOV value.
Auto Shoot	Auto Shoot	auto_shoot	true/false	false	Enables automatic shooting behavior.
Input Method	Arduino 16-bit Mouse	arduino_16_bit_mouse	true/false	false	Arduino input only. Sends wider mouse movement values.
Input Method	Arduino Enable Keys	arduino_enable_keys	true/false	false	Arduino input only. Enables keyboard key output through Arduino.

Other GUI-Exposed Config Controls

Section	Control	Config key	Type/options	Current	Notes
Game Profile	Active Game Profile	active_game	profile names	UNIFIED	Selects the active game profile.
Manage Profiles	Game profile rows	Games.<profile>	name = sens,yaw,pitch[,true ,baseFOV]	profile/local	Custom profiles can be added/deleted; UNIFIED is read-only.
Input Method	Mouse Input Method	input_method	WIN32, GHUB, RAZER, ARDUINO, KMBOX_NET, KMBOX_A, MAKCU	RAZER	Changes active mouse backend.
Input Method	Arduino Port	arduino_port	COM1-COM30	COM0	Arduino input only.
Input Method	Arduino Baudrate	arduino_baudrate	9600-115200	115200	Arduino input only.
Input Method	Kmbox Net IP	kmbox_net_ip	text	10.42.42.42	Saved with Save & Reconnect.
Input Method	Kmbox Net Port	kmbox_net_port	text	1984	Saved with Save & Reconnect.
Input Method	Kmbox Net UUID	kmbox_net_uuid	text	DEADC0DE	Saved with Save & Reconnect.
Input Method	Kmbox A PIDVID	kmbox_a_pidvid	PPPPVVVV		Saved with Save & Reconnect.
Input Method	Makcu Port	makcu_port	AUTO, COM1-COM30	AUTO	Makcu input only.
Input Method	Makcu Baudrate	makcu_baudrate	115200-4000000	4000000	Makcu input only.

Neural Tab

Sliders

Section	Slider	Config key	Range	Current	Notes
Neural Tracker	Association blend	neural_tracker_blend	0.00-1.00	1.000	Blend strength between regular and neural association output.

Activate/Deactivate

Section	Control	Config key	Values	Current	Notes
Neural Tracker	Enable neural association	neural_tracker_enabled	true/false	true	Enables learned association/tracking helper logic.

Other GUI-Exposed Config Controls

Section	Control	Config key	Type/options	Current	Notes
Neural Tracker	Association model	neural_tracker_model_path	path	training/models/neural_tracker.onnx	Path to the neural tracker ONNX file.

AI Tab

Sliders

Section	Slider	Config key	Range	Current	Notes
Detection	Confidence Threshold	confidence_threshold	0.01-1.00	0.10	Minimum detector confidence accepted as a detection.
Detection	NMS Threshold	nms_threshold	0.00-1.00	0.50	Non-max suppression overlap threshold.
Detection	Max Detections	max_detections	1-100	100	Maximum detections retained after model output processing.
Depth Runtime	Depth FPS	depth_fps	0-120	100	CUDA builds only. 0 disables depth debug update throttling path.
Depth Runtime	Depth Mask FPS	depth_mask_fps	1-30	5	CUDA builds only. Mask update frequency.
Depth Mask	Depth Mask Near %	depth_mask_near_percent	1-100	20	CUDA builds only. Percentile cutoff for the near-depth mask.
Depth Mask	Depth Mask Expand (px)	depth_mask_expand	0-128	0	CUDA builds only. Expands the generated mask in pixels.
Depth Mask	Depth Mask Hold Frames	depth_mask_hold_frames	0-120	0	CUDA builds only. Reuses a recent mask for this many detector frames.
Depth Mask	Depth Mask Alpha	depth_mask_alpha	0-255	90	CUDA builds only. Visual alpha for the depth mask overlay.

Activate/Deactivate

Section	Control	Config key	Values	Current	Notes
Depth Inference	Enable Depth Inference	depth_inference_enabled	true/false	true	CUDA builds only. DML build shows a requires-CUDA message.
Depth Mask	Enable Depth Mask	depth_mask_enabled	true/false	false	CUDA builds only. Masks detections/capture based on estimated depth.
Depth Mask	Depth Mask Invert	depth_mask_invert	true/false	false	CUDA builds only. Flips near/far mask selection.
Depth Mask	Depth Debug Overlay (Game)	depth_debug_overlay_enabled	true/false	false	CUDA builds only. Shows depth debug output in the game overlay.

Other GUI-Exposed Config Controls

Section	Control	Config key	Type/options	Current	Notes
Model	Model	ai_model	models folder	sunxds_0.8.2_D ML.onnx	Selects detector model from available files.
Backend	Backend	backend	TRT, DML	DML	CUDA builds only; DML build does not expose this combo.
Depth Inference	Depth model	depth_model_path	models/depth	depth_anything_ v2.engine	CUDA builds only.
Depth Mask	Depth colormap	depth_colormap	0-21. OpenCV colormap index	18	CUDA builds only.

Buttons Tab

Sliders

No saved GUI sliders in this tab.

Activate/Deactivate

Section	Control	Config key	Values	Current	Notes
Arrow Key Options	Enable arrows keys options	enable_arrows_settings	true/false	false	Allows arrow-key adjustment of target offsets.

Other GUI-Exposed Config Controls

Section	Control	Config key	Type/options	Current	Notes
Button Bindings	Targeting Buttons	button_targeting	key list	RightMouseButton	Supports multiple bindings with + and -.
Button Bindings	Shoot Buttons	button_shoot	key list	LeftMouseButton	Supports multiple bindings with + and -.
Button Bindings	Zoom Buttons	button_zoom	key list	RightMouseButton	Supports multiple bindings with + and -.
Button Bindings	Exit Buttons	button_exit	key list	F2	Supports multiple bindings with + and -.
Button Bindings	Pause Buttons	button_pause	key list	F3	Supports multiple bindings with + and -.
Button Bindings	Reload Config Buttons	button_reload_config	key list	F4	Supports multiple bindings with + and -.
Button Bindings	Overlay Buttons	button_open_overlay	key list	Home	Supports multiple bindings with + and -.

Overlay Tab

Sliders

Section	Slider	Config key	Range	Current	Notes
Visual	Overlay Opacity	overlay_opacity	220-255	225	Window opacity. The UI clamps to a readable minimum of 220.
Visual	UI Fine Scale	overlay_ui_scale	0.85-1.35	1.00	Fine-grained scaling for the overlay UI.

Activate/Deactivate

Section	Control	Config key	Values	Current	Notes
Capture Privacy	Hide Overlays From Recording	overlay_exclude_from_capture	true/false	true	Applies capture exclusion so overlay windows are hidden from supported recording/capture paths.

Other GUI-Exposed Config Controls

Section	Control	Config key	Type/options	Current	Notes
Config	Save Config	config.ini	writes config.ini	n/a	Immediately writes the current runtime config.
Config	Load Config	config.ini	reads config.ini	n/a	Merges config.ini into current runtime values and refreshes runtime-sensitive paths.

Game Overlay Tab

Sliders

Section	Slider	Config key	Range	Current	Notes
General	Overlay Max FPS (0 = uncapped)	game_overlay_max_fps	0-256	0	Caps game overlay render rate. 0 is uncapped.
Box Color	A	game_overlay_box_a	0-255	255	Detection box alpha.
Box Color	R	game_overlay_box_r	0-255	0	Detection box red channel.
Box Color	G	game_overlay_box_g	0-255	255	Detection box green channel.
Box Color	B	game_overlay_box_b	0-255	0	Detection box blue channel.
Box Color	Box Thickness	game_overlay_box_thickness	0.5-10.0	2.00	Detection box line thickness.
Capture Frame	A	game_overlay_frame_a	0-255	180	Capture frame alpha.
Capture Frame	R	game_overlay_frame_r	0-255	255	Capture frame red channel.
Capture Frame	G	game_overlay_frame_g	0-255	255	Capture frame green channel.
Capture Frame	B	game_overlay_frame_b	0-255	255	Capture frame blue channel.
Capture Frame	Frame Thickness	game_overlay_frame_thickness	0.5-10.0	1.50	Capture frame line thickness.
Future Point Style	Point Radius	game_overlay_future_point_radius	1.0-20.0	5.00	Radius of future-position points.
Future Point Style	Point Step Alpha Falloff	game_overlay_future_alpha_falloff	0.10-5.00	1.00	Controls how quickly future-point alpha fades along the path.
Icon Overlay	Icon Width	game_overlay_icon_width	4-512	64	Icon width in pixels.
Icon Overlay	Icon Height	game_overlay_icon_height	4-512	64	Icon height in pixels.
Icon Overlay	Icon Offset X	game_overlay_icon_offset_x	-500.0-500.0	0.00	Horizontal icon offset from the selected anchor.
Icon Overlay	Icon Offset Y	game_overlay_icon_offset_y	-500.0-500.0	0.00	Vertical icon offset from the selected anchor.
Aim Simulation	Sim X	aim_sim_x	-3000-3000	24	Aim simulation window X position.
Aim Simulation	Sim Y	aim_sim_y	-3000-3000	24	Aim simulation window Y position.
Aim Simulation	Sim Width	aim_sim_width	220-1600	560	Aim simulation window width.
Aim Simulation	Sim Height	aim_sim_height	180-1000	360	Aim simulation window height.
Aim Simulation	Sim FPS Min	aim_sim_fps_min	15-360	90	Minimum simulated frame rate.
Aim Simulation	Sim FPS Max	aim_sim_fps_max	15-360	120	Maximum simulated frame rate. The GUI keeps min and max ordered.
Aim Simulation	FPS Jitter	aim_sim_fps_jitter	0.000-0.800	0.150	Randomized FPS variance for the simulation.
Aim Simulation	Capture Delay (ms)	aim_sim_capture_delay_ms	0.0-80.0	6.00	Simulated capture delay.
Aim Simulation	Inference Delay (ms)	aim_sim_inference_delay_ms	0.0-120.0	12.00	Manual simulated inference delay when not using live timing.
Aim Simulation	Input Delay (ms)	aim_sim_input_delay_ms	0.0-60.0	2.00	Simulated input delay.
Aim Simulation	Extra Delay (ms)	aim_sim_extra_delay_ms	0.0-60.0	2.00	Additional simulated processing delay.
Aim Simulation	Target Max Speed	aim_sim_target_max_speed	20-2500	560.00	Maximum simulated target speed.
Aim Simulation	Target Accel	aim_sim_target_accel	20-10000	1850.00	Simulated target acceleration.
Aim Simulation	Target Stop Chance	aim_sim_target_stop_chance	0.00-0.95	0.25	Probability that the simulated target pauses during a retarget.

Activate/Deactivate

Section	Control	Config key	Values	Current	Notes
General	Enable	game_overlay_enabled	true/false	false	Enables the separate game overlay renderer.
General	Draw Detection Boxes	game_overlay_draw_boxes	true/false	true	Shows detection boxes in the game overlay.
General	Draw Future Positions	game_overlay_draw_future	true/false	true	Shows predicted future target points.
General	Draw Wind Debug Tail	game_overlay_draw_wind_tail	true/false	true	Shows the wind-mouse debug trail when wind motion is active.
General	Show Target Correction	game_overlay_show_target_correction	true/false	true	Shows correction indicators for target prediction/aim adjustment.
Capture Frame	Draw Capture Frame	game_overlay_draw_frame	true/false	true	Draws the capture frame rectangle.
Icon Overlay	Enable Icon Overlay	game_overlay_icon_enabled	true/false	false	Enables drawing an icon relative to detections.

Section	Control	Config key	Values	Current	Notes
Aim Simulation	Enable Aim Simulation Window	aim_sim_enabled	true/false	false	Shows the aim simulation overlay window.
Aim Simulation	Use Live Inference Delay	aim_sim_use_live_inference	true/false	true	Uses live backend timing for inference delay instead of the manual value.
Aim Simulation	Show Delayed Observation	aim_sim_show_observed	true/false	true	Shows delayed target observation in the simulation.
Aim Simulation	Show Trajectory History	aim_sim_show_history	true/false	true	Shows historical target/aim trails.
Aim Simulation	Show Kalman Debug	aim_sim_show_kalman_debug	true/false	true	Shows Kalman-related debug visuals in the simulation.

Other GUI-Exposed Config Controls

Section	Control	Config key	Type/options	Current	Notes
Icon Overlay	Icon Path	game_overlay_icon_path	path	icon.png	Image path for icon overlay.
Icon Overlay	Icon Class (-1 = all)	game_overlay_icon_class	-1 or class id	-1	Restricts icon to one class when ≥ 0 .
Icon Overlay	Icon Anchor	game_overlay_icon_anchor	center, top, bottom, head	center	Anchor point used for icon placement.

Stats Tab

The Stats tab is read-only in the current GUI. It displays timing graphs, capture FPS, capture details, and CUDA/depth status where available.

Debug Tab

Sliders

No saved GUI sliders in this tab.

Activate/Deactivate

Section	Control	Config key	Values	Current	Notes
Screenshot Buttons	Verbose console output	verbose	true/false	false	Enables extra console logging.
Log File	Enable log file	debug_log_file_enabled	true/false	false	Writes logs to debug_log_file_path when enabled.
Neural Diagnostics	Log neural tracker associations	neural_tracker_log_enabled	true/false	false	Writes neural association diagnostics to neural_tracker_log_path.
Neural Diagnostics	Show neural tracker debug	neural_tracker_debug_enabled	true/false	false	Shows neural tracker debug output in the game overlay path.

Other GUI-Exposed Config Controls

Section	Control	Config key	Type/options	Current	Notes
Screenshot Buttons	Screenshot buttons	screenshot_button	key list	None	Supports multiple bindings with + and -.
Screenshot Buttons	Screenshot delay	screenshot_delay	step 50/500	500	Delay before screenshot capture.
Log File	Log file path	debug_log_file_path	path	logs/OBS.log	Used when Enable log file is on.
Neural Diagnostics	Neural tracker log	neural_tracker_log_path	path	training/logs/neural_tracker_association.csv	CSV path for neural association diagnostics.

Config.ini Settings Not In The GUI

These settings are editable by changing config.ini directly. Some are build-gated: CUDA rows are emitted only by CUDA builds, and depth rows are GUI-editable only when the app is compiled with USE_CUDA.

Section	Setting	Current/default	Range/options	Visibility	Guidance	Notes
Mouse prediction	prediction_futurePositions	20	integer	Config only	Number of future target positions to retain/use for prediction visualization.	Saved in current config.ini, but no direct GUI control exists.
Mouse prediction	draw_futurePositions	true	true/false	Config only	Enables drawing future positions in preview/debug paths.	Separate from game_overlay_draw_future.
PID governor	pid_governor_model_path	training/models/pid_governor.onnx	path	Config only	Model path for the ONNX PID governor.	Relative paths are resolved against current/execute parent locations.
AI backend	backend	DML	DML or TRT	DML config-only / CUDA GUI	Selects DirectML or TensorRT backend.	GUI combo exists only in CUDA builds.
AI backend	dml_device_id	0	integer adapter id	Config only	DirectML adapter index used by ONNX Runtime DML.	Not exposed in the GUI.
System reserves	cpuCoreReserveCount	4	integer	Config only	CPU cores reserved away from worker assignment.	Not exposed in the GUI.
System reserves	systemMemoryReserveMB	2048	MB	Config only	System memory reserve used by runtime resource planning.	Not exposed in the GUI.
Custom classes	class_player	0	integer class id	Config only	Detector class id treated as player/body.	Not exposed in the GUI.
Custom classes	class_head	1	integer class id	Config only	Detector class id treated as head.	Not exposed in the GUI.
Debug	show_fps	false	true/false	Config only	Legacy/debug FPS display flag.	Saved in config.ini but not exposed in the current GUI.
Game profiles	Games.UNIFIED	1.00,0.02,0.02	sens.yaw,pitch[,true,baseFOV]	Config only	Default profile row used for degree-to-count conversion.	UNIFIED is shown read-only in the GUI.
Depth	depth_inference_enabled	true	true/false	DML config-only / CUDA GUI	Enables depth inference runtime.	In the DML build this remains config-only; CUDA builds expose it in AI -> Depth.
Depth	depth_model_path	depth_anything_v2.engine	path	DML config-only / CUDA GUI	Depth model or engine path.	In the DML build this remains config-only; CUDA builds expose it in AI -> Depth.
Depth	depth_fps	100	0-120	DML config-only / CUDA GUI	Depth debug inference rate.	In the DML build this remains config-only; CUDA builds expose it in AI -> Depth.
Depth	depth_colormap	18	0-21	DML config-only / CUDA GUI	OpenCV colormap index for depth debug view.	In the DML build this remains config-only; CUDA builds expose it in AI -> Depth.
Depth	depth_mask_enabled	false	true/false	DML config-only / CUDA GUI	Enables depth mask generation.	In the DML build this remains config-only; CUDA builds expose it in AI -> Depth.
Depth	depth_mask_fps	5	1-30 GUI, >=0 load clamp	DML config-only / CUDA GUI	Depth mask update frequency.	In the DML build this remains config-only; CUDA builds expose it in AI -> Depth.
Depth	depth_mask_near_percent	20	1-100	DML config-only / CUDA GUI	Near-depth percentile threshold for mask generation.	In the DML build this remains config-only; CUDA builds expose it in AI -> Depth.
Depth	depth_mask_expand	0	0-128 px	DML config-only / CUDA GUI	Expands the mask after generation.	In the DML build this remains config-only; CUDA builds expose it in AI -> Depth.
Depth	depth_mask_hold_frames	0	0-120	DML config-only / CUDA GUI	Frames to hold/reuse the last mask.	In the DML build this remains config-only; CUDA builds expose it in AI -> Depth.
Depth	depth_mask_alpha	90	0-255	DML config-only / CUDA GUI	Depth mask overlay alpha.	In the DML build this remains config-only; CUDA builds expose it in AI -> Depth.
Depth	depth_mask_invert	false	true/false	DML config-only / CUDA GUI	Inverts near/far mask selection.	In the DML build this remains config-only; CUDA builds expose it in AI -> Depth.
Depth	depth_debug_overlay_enabled	false	true/false	DML config-only / CUDA GUI	Shows depth debug overlay in the game overlay.	In the DML build this remains config-only; CUDA builds expose it in AI -> Depth.
Advanced mouse	minSpeedMultiplier	0.1	float	Loadable hidden key	Legacy minimum movement speed multiplier.	Loadable from config.ini if added manually; saveConfig does not currently emit it.
Advanced mouse	maxSpeedMultiplier	0.1	float	Loadable hidden key	Legacy maximum movement speed multiplier.	Loadable from config.ini if added manually; saveConfig does not currently emit it.

Section	Setting	Current/default	Range/options	Visibility	Guidance	Notes
Advanced mouse	predictionInterval	0.01	seconds	Loadable hidden key	Prediction sample interval used by legacy/auxiliary prediction logic.	Loadable from config.ini if added manually; saveConfig does not currently emit it.
Legacy snap	snapRadius	1.5	float	Config only	Legacy snap radius.	Loadable hidden key; no GUI control.
Legacy snap	nearRadius	25.0	float	Config only	Legacy near-target radius.	Loadable hidden key; no GUI control.
Legacy snap	speedCurveExponent	3.0	float	Config only	Legacy speed curve exponent.	Loadable hidden key; no GUI control.
Legacy snap	snapBoostFactor	1.15	float	Config only	Legacy snap boost multiplier.	Loadable hidden key; no GUI control.
Legacy recoil	easynorecoil	false	true/false	Config only	Legacy recoil helper flag.	Loadable hidden key; no GUI control and not emitted by saveConfig.
Legacy recoil	easynorecoilstrength	0.0	float	Config only	Legacy recoil helper strength.	Loadable hidden key; no GUI control and not emitted by saveConfig.
Wind mouse	wind_mouse_enabled	false	true/false	Config only	Enables wind-style motion noise in supported runtime/simulation paths.	Loadable hidden key; no GUI control.
Wind mouse	wind_G	18.0	float	Config only	Wind mouse gravity parameter.	Loadable hidden key; no GUI control.
Wind mouse	wind_W	15.0	float	Config only	Wind mouse wind parameter.	Loadable hidden key; no GUI control.
Wind mouse	wind_M	10.0	float	Config only	Wind mouse max-step parameter.	Loadable hidden key; no GUI control.
Wind mouse	wind_D	8.0	float	Config only	Wind mouse damping/distance parameter.	Loadable hidden key; no GUI control.
CUDA	export_enable_fp8	false	true/false	CUDA config-only	Enables FP8 TensorRT export where supported.	CUDA builds only; no GUI checkbox.
CUDA	export_enable_fp16	true	true/false	CUDA config-only	Enables FP16 TensorRT export where supported.	CUDA builds only; no GUI checkbox.
CUDA	use_cuda_graph	false	true/false	CUDA config-only	Enables CUDA graph execution path.	CUDA builds only; no GUI checkbox.
CUDA	use_pinned_memory	true	true/false	CUDA config-only	Uses pinned host memory for CUDA paths.	CUDA builds only; no GUI checkbox.
CUDA	gpuMemoryReserveMB	2048	MB	CUDA config-only	GPU memory reserve for CUDA resource planning.	CUDA builds only; no GUI control.
CUDA	enableGpuExclusiveMode	true	true/false	CUDA config-only	Attempts GPU exclusive mode at startup.	CUDA builds only; no GUI checkbox.

Source Map

Area	Source
Tab order	overlay/overlay.cpp
Capture	overlay/draw_capture.cpp and overlay/draw_debug.cpp
Target	overlay/draw_target.cpp
Mouse	overlay/draw_mouse.cpp
Neural	overlay/draw_neural.cpp
AI and Depth	overlay/draw_ai.cpp and overlay/draw_depth.cpp
Buttons	overlay/draw_buttons.cpp
Overlay	overlay/draw_overlay.cpp
Game Overlay	overlay/draw_game_overlay.cpp
Debug	overlay/draw_debug.cpp
Config load/save/defaults	config/config.cpp and config/config.h